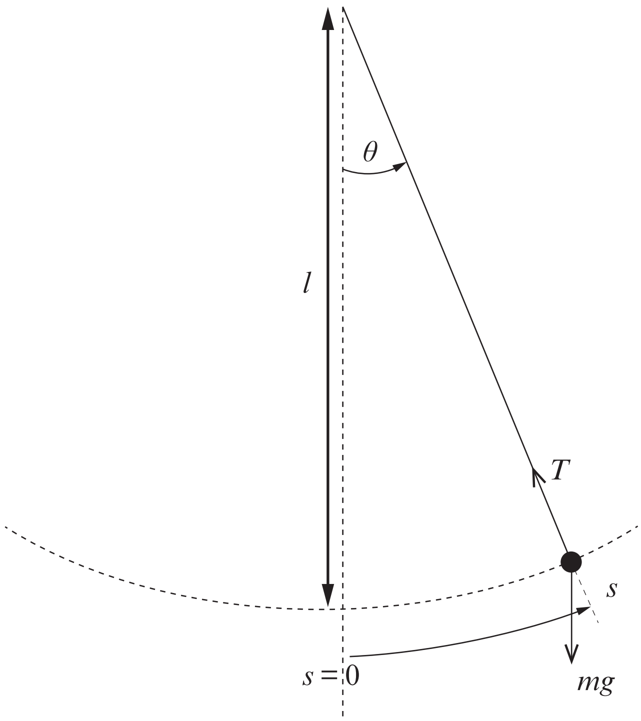
Examiner only

6. *(a)* The pendulum in the figure below has a small bob of mass, m, suspended at the bottom of a light string of length, l. The string is shown at an angle, θ, to the vertical and the mass, m, is at a distance, s, along the arc. The forces acting on the mass are shown.



(i) Name the **two** forces acting on the mass. [1]

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(ii) By considering these forces show that:

resultant force component on the mass along the arc = $-mg\sin\theta$

You may add to the diagram if you wish. [3]

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Examiner only

(iii) If the oscillation is small so that $\sin\theta \simeq \theta$ show in clear steps that the acceleration along the arc may be written as: [2]

$$\text{acceleration} = -\frac{gs}{l}$$

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(iv) Discuss whether the equation in part *(a)*(iii) satisfies the definition of simple harmonic motion. [2]

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(b) A small mass oscillates at the lower end of a pendulum of length 1.20 m.

(i) Determine its:

I. period; [2]

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II. frequency. [1]

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(ii) If the maximum displacement angle of the mass is 0.067 rad, justify the use of simple harmonic motion in part *(b)*(i). [2]

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